



NAMIBIA UNIVERSITY  
OF SCIENCE AND TECHNOLOGY

**HCI711S**  
**GROUP ASSIGNMENT 2026**  
**ADAPTIVE LEARNING INTERFACES:**

*Functionality, Advantages, and Cybersecurity Implications*

Group 4

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**Introduction**

Human-Computer Interaction (HCI) has changed the way people use technology. In the past, computers mainly used command-line systems, but today technology has become more advanced with AI-powered and personalised systems. One major development is Adaptive Learning Interfaces (ALIs). These systems use artificial intelligence, machine learning, and real-time data to adjust digital experiences to suit each user's needs and preferences (Stephanidis et al., 2025). This topic shows how HCI continues to grow by combining AI with new trends in technology.

This paper discusses Adaptive Learning Interfaces, including how they work, real-world examples, their benefits, and the cybersecurity risks they may create. It also looks at important HCI topics such as ethics, privacy, usability, intellectual property, laws, accuracy, and the needs of different users, using information from current research and industry sources.

## **Understanding the User in Adaptive Learning Interfaces**

Understanding the user is one of the most important parts of an Adaptive Learning Interface (ALI). These systems constantly collect information such as how users interact with the system, how quickly they respond, the mistakes they make, their preferences, and sometimes even physical or emotional conditions. This information helps the system adjust and personalise the learning experience (Daily News, 2023). User-centred design (UCD) explains that people are different in many ways, including their knowledge, thinking ability, culture, age, and accessibility needs. Because of this, a good ALI should be designed to support all types of users fairly and effectively.

It is also important for ALIs to understand the user's situation or environment. For example, a person using a mobile phone while travelling may have different needs and limitations compared to someone using a desktop computer at home or work. If an ALI does not consider these differences, it may provide content at the wrong time, in the wrong format, or at the wrong difficulty level, which can reduce how effective the system is (Stephanidis et al., 2025).

## **Adaptive Learning Interfaces: Definition and Functional Examples**

Adaptive Learning Interfaces (ALIs) are digital systems that change and adjust according to each user's needs. Unlike traditional systems that give the same experience to everyone, ALIs personalise learning by analysing user information in real time. They mainly work through four important parts:

- **User Model:** keeps track of the user's knowledge, learning style, preferences, and progress.
- **Adaptive Engine:** uses AI and machine learning to study the user's information and decide what changes should be made.

- **Content Repository:** stores learning materials and interface features that can be arranged differently for each user.
- **Feedback Mechanism:** gives instant feedback to help users learn, fix mistakes, and improve their understanding.

Many popular platforms already use these features in practice:

- Duolingo uses adaptive difficulty and spaced repetition to help users remember vocabulary. The system monitors which words users struggle with and increases practice for those areas based on memory patterns (Daily News, 2023).
- Khan Academy uses a mastery-based learning system where students must fully understand a topic before moving forward. The platform recommends activities based on the learner's weak areas.
- Knewton, now part of Wiley, uses large amounts of learner data to predict the best content for each student and improve learning success.
- Microsoft Immersive Reader adjusts text size, spacing, and reading support features according to a user's accessibility needs, showing how ALIs can also help people with different learning or reading challenges.

## **Advantages of Adaptive Learning Interfaces**

Adaptive Learning Interfaces (ALIs) offer many benefits for both learners and educators. One of the biggest advantages is personalisation. Instead of giving every learner the same content, ALIs adjust lessons to match each person's needs and learning progress. This helps reduce confusion and keeps users more engaged because they spend time focusing on areas they still need to improve rather than repeating what they already know (Stephanidis et al., 2025).

ALIs also improve learning results. Studies show that methods such as spaced repetition, step-by-step mastery learning, and instant personalised feedback help people remember information better over a longer period compared to traditional learning systems.

Another important benefit is accessibility and inclusivity. ALIs can support people with different abilities, languages, and literacy levels by offering features such as text-to-speech, simpler language options, and adjustable display settings. This helps make digital learning fairer and more accessible to everyone.

In addition, ALIs provide real-time assessment. Instead of relying only on major tests or exams, these systems continuously check a learner's progress through smaller activities and feedback during the learning process. This gives both learners and teachers quick information about areas that need improvement.

ALIs are also highly scalable because they can provide personalised learning to many users at the same time without needing large increases in staff or resources. AI-powered systems

can even identify learners who may be struggling early on, allowing support to be given before the learner fails or falls behind (Daily News, 2023).

## **Cybersecurity Risks and Mitigations**

Despite their benefits, ALIs introduce serious cybersecurity risks, heightened by the sensitivity of the data they collect and the critical environments in which they operate.

### **Security Concern 1 – Data Privacy and Unauthorised Data Collection**

Adaptive Learning Interfaces (ALIs) collect a large amount of personal information from users. This can include learning behaviour, academic results, thinking patterns, and sometimes even physical or health-related data. Over time, this information creates detailed profiles about individuals, which can become attractive targets for cybercriminals. If security systems fail because of hacking, insider threats, or weak third-party systems, personal data could be exposed and used for identity theft, unfair treatment, or manipulation of users (Stephanidis et al., 2025).

To reduce these risks, organisations should only collect information that is truly necessary. Strong security measures such as encryption technologies like AES-256 and TLS 1.3 should be used to protect data. Access to information should also be limited through role-based access control (RBAC), where only authorised people can view certain data. In addition, regular privacy impact assessments (PIAs) should be carried out, and systems must follow privacy and cybersecurity laws such as GDPR, COPPA, and Namibia's Electronic Transactions and Cybercrime Act 42 of 2019.

### **Security Concern 2 – Adversarial Attacks on AI/ML Models**

The adaptive engines powering ALIs are vulnerable to adversarial attacks — deliberate manipulation of input data to cause incorrect model outputs. An attacker could manipulate interaction data to game the system, artificially inflating assessed competency. More broadly, data poisoning attacks, where malicious data is injected into training pipelines, can degrade recommendation quality for all users, often without detection for extended periods.

Mitigations include input validation and anomaly detection to flag suspicious interaction patterns, adversarial training techniques that expose models to attack examples during development, Explainable AI (XAI) frameworks for auditing adaptive decisions and detecting model drift, and cryptographic signing of training data to protect ML pipeline integrity.

### **Security Concern 3 – Authentication and Authorisation Vulnerabilities**

Web-based and cloud-based Adaptive Learning Interfaces (ALIs) can have security weaknesses if they are not properly protected. One common issue is weak login security, such as not using multi-factor authentication (MFA), unsafe session login tokens, or poor role-based access control (RBAC). These weaknesses can allow attackers to take over user accounts or gain higher-level access than they should have. If a student's account is hacked, it can lead to cheating or academic fraud. If a teacher's or administrator's account is compromised, it can expose large amounts of student performance data. A growing problem is "credential stuffing," where attackers use leaked usernames and passwords from other websites to automatically try and break into accounts.

To reduce these risks, systems should require MFA for everyone. They should also use a zero-trust approach, meaning every login attempt is checked, even if it comes from a trusted network. Access should follow the "least privilege" rule, where users only get the permissions, they truly need, and these permissions should be reviewed regularly. In addition, session tokens should be randomly generated, expire quickly, and systems should limit repeated login attempts to block automated attacks.

## **HCI Guidelines and Ethical Considerations**

Adaptive Learning Interfaces (ALIs) must be designed with important responsibilities in mind.

**Ethics:** ALIs should make their decisions clear and explainable so that they can be checked and understood. One major risk is algorithm bias, where the system learns unfair patterns from old data and unintentionally disadvantages certain groups of people. To prevent this, systems need regular bias checks, more diverse training data, and ways for users to challenge or question the system's decisions (Stephanidis et al., 2025).

**Privacy:** Users must clearly understand what data is being collected, why it is collected, how it is used, and who can access it. This is called informed consent. Privacy should be built into the system from the beginning, not added later. Laws like GDPR also give users the right to see, change, or delete their personal data.

**Usability:** Even though ALIs use advanced technology, they should still be simple and easy to use. People with different levels of technical skill should be able to use them without confusion. The system should not change too quickly or in a way that confuses users. Regular testing with different types of users is important to keep the design user-friendly.

**Intellectual Property:** ALIs often use learning materials from different sources. These materials must be properly licensed and used legally. Even when content is adapted or reshaped for users, copyright rules must still be followed.

Legislation: ALIs must follow all relevant laws, including GDPR, COPPA, and Namibia's Electronic Transactions and Cybercrime Act 42 of 2019. These laws are not optional and must be fully respected.

Accuracy, User Diversity, and Context: ALIs need to be checked regularly to make sure their recommendations stay correct and up to date. They should be designed for all kinds of users, including different languages, abilities, and devices. The system must also understand context, such as where and how the user is learning, to make sure the experience is always appropriate and helpful.

## **Conclusion**

Adaptive Learning Interfaces (ALIs) are one of the biggest improvements in Human-Computer Interaction. They use artificial intelligence and real-time data to create learning experiences that are more personal, inclusive, and able to work on a large scale. These systems are used in areas like education, workplaces, healthcare, and accessibility. Because they can predict user needs, they are becoming an important part of the future of HCI.

However, ALIs also come with serious risks. Since they rely on large amounts of personal data, they can be vulnerable to problems such as data breaches, attacks on AI systems, and weak login security. These issues need strong technical protection and proper management rules to keep users safe.

To make ALIs trustworthy, they must follow important HCI principles from the beginning. This includes being transparent, fair, protecting privacy by design, and following legal requirements. As Stephanidis et al. (2025) explain, the combination of AI, personalisation, and security is shaping the future of HCI. Because of this, understanding the cybersecurity risks of ALIs is very important.

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